

Q1. Derivative in Gradient Descent

- **Formula:** $\theta_1 := \theta_1 - \alpha \frac{\partial J}{\partial \theta_1}$
- If $\frac{\partial J}{\partial \theta_1} > 0$, subtracting a positive quantity decreases θ_1 .
- **Listed Answer:** A. θ_1 decreases — **Correct**

Q2. Overfitting Definition

- Overfitting occurs when the model fits the training data too closely, resulting in very low training error but high test/validation error.
- **Listed Answer:** A. Low training error, high test error — **Correct**

Q3. Dimensionality Reduction

- PCA (Principal Component Analysis) reduces the number of features while retaining maximum variance/information.
- **Listed Answer:** B. Principal Component Analysis (PCA) — **Correct**

Q4. Standardized Z-Score Calculation

- **Formula:** $Z = \frac{X - \mu}{\sigma} = \frac{73,000 - 54,000}{16,000} = \frac{19,000}{16,000} = 1.1875$
- **Listed Answer:** c. 1.1875 — **Correct**

Q5. 3-Sigma Outlier Upper Bound

- **Formula:** $\text{Upper Bound} = \mu + 3\sigma = 50 + 3(5) = 65$
- **Listed Answer:** C. 65 — **Correct**

Q6. Model-Based Learning

- Model-based learning detects patterns in training data to construct an explicit predictive model.
- **Listed Answer:** D. Building a predictive model by identifying patterns in data — **Correct**

Q7. Deep Learning Optimizer Mathematical Concept

- Gradients (slope of the loss function) guide weight updates in gradient descent/backpropagation.
- **Listed Answer:** D. Derivatives and gradients — **Correct**

Q8. IQR Outlier Upper Bound

- **Calculation:**

$$\text{IQR} = Q_3 - Q_1 = 20 - 10 = 10$$

$$\text{Upper Bound} = Q_3 + 1.5 \times \text{IQR} = 20 + 1.5(10) = 35$$

- **Listed Answer:** A. 35 — **Correct**

Q9. Instance-Based Learning

- Instance-based algorithms compare new observations directly against memorized training instances (e.g., k-NN).

- **Listed Answer:** D. It compares new instances with previously observed examples — **Correct**

Q10. Random Subsampling Split

- **Calculation:** $\frac{2}{3} \times 150 = 100$
- **Listed Answer:** B. 100 — **Correct**

Every answer in your list is 100% correct.